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Palaeomagnetism of the Egersund–Ogna anorthosite, Rogaland, Norway, and the position of Fennoscandia in the Late Proterozoic

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SUMMARY

The 930-Ma Egersund–Ogna anorthosite is part of the Rogaland igneous complex of massif-type anorthosites intruded into Sveconorwegian basement of southwestern Norway. Average susceptibilities on samples from 13 sites range from 0.03 to 2.24 $\times$ 10^{-3} SI and natural remanent intensities range from 0.004 to 1.54 A m^{-1}. Corresponding Koenigsberger ratios range from 3 to 148 with a mean value of 36, indicating that magnetic field surveys will observe remanent-controlled magnetic anomalies. Optical observations indicate that haemo-ilmenite is the major oxide phase present, with lesser amounts of ilmeno-haematite and rare magnetite. Remanent directions from all samples are characterized by steeply negative inclinations with variable northwest declinations. Thermal demagnetization reveals square-shouldered demagnetization curves with little or no loss of intensity until 550 or 575 °C. Alternating field demagnetization produces a wide range of demagnetization behaviours with mean destructive fields varying from less than 5 mT to greater than 80 mT. There is little evidence for overprinting or secondary components, and all information points to a remanence acquired during initial cooling of the anorthosite at $\sim$ 900 Ma. Mean directional data for the 13 sites are $D = 325.^{\circ}9$, $I = -80.^{\circ}1$ and $a_{95} = 4.^{\circ}9$. The magnetic pole calculated for Egersund–Ogna is at 42.°1S latitude and 200.°4 E longitude ($A_{95} = 9.^{\circ}0$), in excellent agreement with an earlier pole determined from Egersund, and similar to other poles from Rogaland igneous complex rocks. This work supports Rodinia reconstructions that place Baltica at high (southern) latitude, $\sim 70^{\circ}$, at about 900 Ma. Apparent polar wander paths for Baltica at this time are ambiguous and it is difficult to discriminate between proposed clockwise or counter-clockwise loops.

Key words: anorthosite, Baltica, haemo-ilmenite, palaeomagnetism, Proterozoic, Rodinia.

1 INTRODUCTION

The task of configuring continental landmasses during the Proterozoic (2700–600 Ma) is a speculative business fuelled by input from numerous geological and geophysical observations. Palaeomagnetic data from well-dated rocks are important for the latitudinal placement of continents and the determination of apparent polar wander paths (APWPs). Although the possibility of one long-lived megacontinent has been suggested (Piper 1987), much attention during the 1990s focused on the possibility of a supercontinent, Rodinia, existing during the Late Proterozoic (Dalziel 1991; Hoffman 1991). Initial suggestions for the configuration of Rodinia were based on orogenic belts of the Grenville orogeny ($\sim$ 1.0 Ga) (Hoffman 1991), still a useful tool (Karlstrom *et al.* 2001), but palaeomagnetic data and resulting APWPs were increasingly important for confirming global alignments and investigating the motion of separate continents (Van der Voo & Meert 1991; Weil *et al.* 1998; Buchan *et al.* 2001). The timing of Rodinia assembly and breakup, as well as the relative position of continental landmasses within Rodinia, is a vital research topic, as seen in recent journal volumes dedicated to the subject (e.g. Meert & Powell 2001). More recently it has been suggested that Rodinia was in flux during much of its $\sim$ 400 Myr history (Torsvik 2003).

Two important crustal blocks make up Rodinia's eastern core, Laurentia (North America including Greenland) and Baltica (Scandinavia, northwestern Russia and Ukraine). The position of Baltica with respect to Laurentia at various times from 1100 to 600 Ma has been suggested on the basis of geological information (Park 1992; Starmer 1996) and palaeomagnetic data (Torsvik *et al.* 1996; Buchan *et al.* 2000, 2001; Hartz & Torsvik 2002). However, the paucity of well-dated palaeomagnetic poles for Baltica allows for various scenarios on its location (Pesonen *et al.* 1989, 1991; Elming *et al.* 1993; Torsvik *et al.* 1996) and resulting APWPs depend on the criteria used for pole selection (Torsvik *et al.* 1996; Weil *et al.* 1998; Buchan *et al.* 2000, 2001; Meert & Torsvik 2003). Although the IAGA Global Paleomagnetic Database contains over

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100 entries for Baltica between 1200 and 500 Ma, most poles are unsuitable for APWP construction. Poles must be omitted both for lack of sufficiently well-determined age information and for lack of adequate sampling to provide reliable geomagnetic poles. These problems, combined with limited or non-existent data over much of the Middle to Late Proterozoic, produce poorly defined and commonly ambiguous APWPs, as discussed recently by Meert & Torsvik (2003).

One of the more ambiguous time periods is right after the Sveconorwegian (= Grenville) orogeny post-1000 Ma but prior to the breakup of Rodinia starting at ∼750 Ma. An excellent place to study Baltica palaeomagnetism is in the Rogaland igneous province, a post-orogenic anorthosite complex intruded into Sveconorwegian pyroxene granulites. Here, pristine rocks of the Tonian period of Neoproterozoic age (930 Ma) with little or no later deformation are known to carry stable remanence (Hargraves & Fish 1972; Poorter 1972; Stearn & Piper 1984). In this paper we undertake a detailed modern palaeomagnetic study on the Egersund–Ogna anorthosite to investigate the position of Baltica in post-Sveconorwegian time.

2 GEOLOGICAL SETTING

The Egersund–Ogna anorthosite body is part of the Rogaland igneous province of southwestern Norway (Fig. 1). The province consists of four large massif anorthosites (Egersund–Ogna, Haland, Helleren and Åna–Sira), the 230 km^2 Bjerkreim–Sokndal layered intrusion, and several smaller intrusive bodies including the Hidra leuconorite and the Farsund charnockite (Marker *et al.* 2003a,b). The bodies intrude into migmatic gneisses that were metamorphosed to granulite facies during the Sveconorwegian orogeny. The anorthosites and associated rocks have been studied for many years (Michot & Michot 1969; Michot 1969) and are currently the subject of extensive geochemical, geochronological and petrologic studies (Duchesne 2001), partly stimulated by the presence of economic haemo-ilmenite ore and building stone deposits (Korneliussen *et al.* 1999).

The Egersund–Ogna body is a dome-shaped massif of anorthosite, composed of over 90 per cent homogeneous plagioclase, with rare megacrysts of orthopyroxene containing exsolved plagioclase and garnet (Duchesne & Maquil 1987). The central part

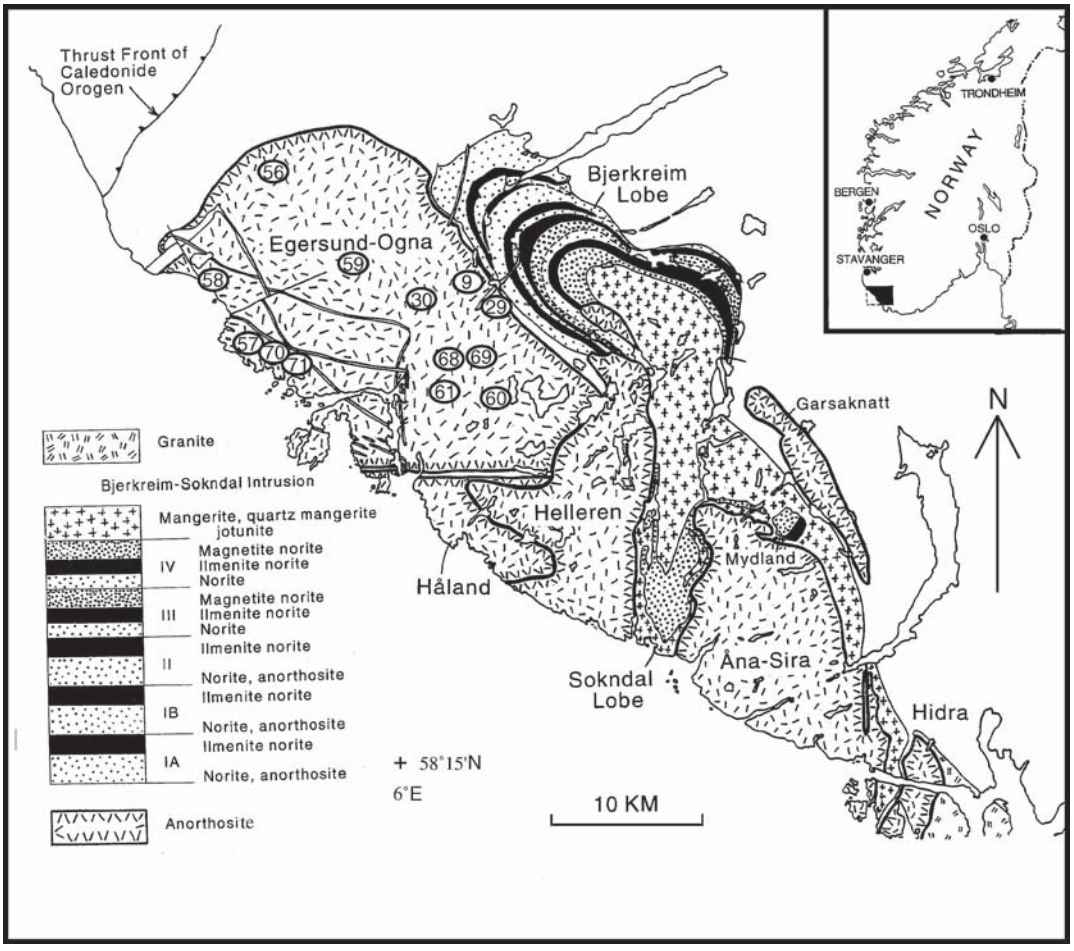


Figure 1. Simplified geological map of the Rogaland igneous province adapted from McEnroe *et al.* (2001b). Shown are the Egersund–Ogna anorthosite, the Haland anorthosite, the Helleren anorthosite, the Åna–Sira anorthosite, the Hidra body, and both the Bjerkreim and Sokndal lobes of the Bjerkreim–Sokndal layered intrusion. Numbered ovals are palaeomagnetic sampling sites in the Egersund–Ogna anorthosite; numbers correspond to sites listed in Table 1. The inset map shows the location of the Rogaland igneous province in southwestern Norway.

of the massif is dominated by granular plagioclase interspersed with plagioclase, and orthopyroxene megacrysts. A narrow marginal zone has variable degrees of foliated anorthosite, involving grain reorientation and recrystallization related to emplacement of the body (Barnichon *et al.* 1999).

Geochronologic studies of the Rogaland igneous province indicate the age of intrusion to be between 930 and 920 Ma, well after the peak metamorphic phase of the Sveconorwegian orogeny (Duchesne 2001). U–Pb determinations on zircons from the Egersund–Ogna anorthosite give mean ages of 929 ± 2 Ma, and 931 ± 3 Ma on megacryst aggregates in the anorthosite, with similar ages for the Åna–Sira (932 ± 3 Ma) and Helleren (932 ± 3 Ma) bodies as well as an age of 931 ± 5 Ma on late-magmatic jotunite dykes (Schärer *et al.* 1996). The Tellnes ore body, intruded into the Åna–Sira anorthosite, gives a slightly younger age of 920 ± 3 Ma (Schärer *et al.* 1996). U–Pb titanite studies on gneisses surrounding the Rogaland igneous province yield an average age of 918 Ma (Bingen & Van Breemen 1996), indicating that the surrounding region cooled through 600 °C at this time.

Although this part of Scandinavia was the site of considerable geological activity in the Precambrian, the Rogaland igneous province appears to be the last major magmatic activity (Marker *et al.* 2003b). The Hunnedalen dyke swarm, a group of several sub-parallel dykes trending northeast–southwest and dated at 855 ± 59 Ma by Sm–Nd whole rock and at 848 ± 27 Ma by $^{40}Ar/^{39}Ar$ on biotite (Walderhaug *et al.* 1999), lie ∼ 50 km to the northeast of the Rogaland complex. The latest activity appears to be the Egersund dykes, a swarm of ESE–WNW trending diabase dykes intruding the anorthosites and surrounding gneiss terrane. U–Pb geochronology gives an age of intrusion of 616 ± 3 Ma (Bingen *et al.* 1998b). The emplacement of these dykes appears to be related to initial breakup of Rodinia in the late Neoproterozoic and the opening of the Iapetus ocean (Torsvik *et al.* 1996; Bingen *et al.* 1998b). A thrust front of the Late Silurian–Early Devonian Caledonide orogen is located approximately 5 km northwest of the northwest margin of the Egersund–Ogna body (Fig. 1), and a greenschist-facies green-biotite isograd extends about 5 km southeast from the front, barely overlapping the edge of the body (Verschure *et al.* 1980; Oftedahl 1980; Bol *et al.* 1995). Beyond the green-biotite isograd there is only a local very weak prehnite–pumpellyite overprint. No later tectonic activity is recorded in Rogaland, and no active faults or deposits younger than Neoproterozoic are mapped in the area (Marker *et al.* 2003a).

3 MAGNETIC MINERALOGY

The bulk magnetic properties of the Egersund–Ogna anorthosite reflect the modal proportions, grain sizes and compositions of the magnetic minerals. Optical observations on polished sections representing each site showed discrete well-crystallized oxide grains from ∼5 to ∼300 μm in diameter. Typical anorthosite samples have approximately 0.5 per cent haemo-ilmenite ± subordinate ilmeno-haematite. Common in all samples are discrete haemo-ilmenite grains with at least two generations of haematite exsolution lamellae (Fig. 2a). Haematite lamellae are observable down to the optical limit of the microscope (∼0.2 μm). Ilmenite hosts may also contain pleonaste lamellae, and sulphide inclusions. Ilmeno-haematite (haematite host with ilmenite exsolution) grains have multiple generations of fine ilmenite exsolution lamellae. Ilmeno-haematite is more abundant in samples from sites 58, 70 and 71. Minor multidomain-size magnetite grains, up to 50 μm in diameter, were observed to be present in a few samples (Fig. 2b). These grains



Figure 2. Photomicrographs of oxides in Egersund–Ogna Anorthosite. (a) Discrete ilmenite grain with exsolution lamellae of haematite (white) and pleonaste (black) in reflected light with a blue filter. The field of view is 30 μm. (b) Large discrete magnetite grain with oxidation exsolution lamellae of ilmenite in reflected light with a blue filter. The field of view is 20 μm. (c) Transmitted-light microphotograph of orthopyroxene with exsolution of ilmenite. Ilmenite contains fine exsolution lamellae of haematite. The field of view is 50 μm.

are relatively free of microstructures, but contain minor {111} oxidation–exsolution lamellae of ilmenite. Samples from site 61 contain large orthopyroxene inclusions. Orthopyroxenes contain coherent rods and blades of ilmenite with exsolved haematite lamellae, where (001) ilmenite is epitaxial on (100) pyroxene (Fig. 2). Discrete small grains of ilmeno-haematite are present in plagioclase grains. The reaction of ilmenite to haematite + rutile ± spinel is locally present in some discrete haemo-ilmenite grains, indicating minor low-temperature oxidation below 400 °C.

Based on detailed studies of exsolution in the haematite–ilmenite series, exsolution lamellae likely continue down to the nanometre scale (McEnroe *et al.* 2001a, 2002; Robinson *et al.* 2002, 2004) and will have a strong effect on the natural remanent magnetization (NRM) of the samples (McEnroe & Brown 2000b). All samples studied here contain either minor amounts of haemo-ilmenite or ilmeno-haematite, with abundant exsolution, and these minerals are considered to be the major carrier of remanence in the Egersund–Ogna anorthosite.

4 PALAEOMAGNETISM

4.1 Sampling

As part of a larger palaeomagnetic study of the Rogaland igneous province, 13 sites were collected around the Egersund–Ogna anorthosite body (Fig. 1). Although several sites are near the disturbed margin, only one site (BK 56) is in foliated rock. From four to seven cores were drilled using a gasoline-powered rock drill and oriented at each location with both sun and magnetic compasses, with one to three specimens being cut from each core.

4.2 Laboratory procedures

NRM measurements on all samples were made on a 2G SQUID magnetometer at the University of Massachusetts. Both alternating field (AF) and thermal demagnetization in incremental steps were performed on suites of specimens from the same cores, using a Molspin AF demagnetizer and an ASC thermal demagnetizer respectively. AF step-wise series went to peak fields of 99 mT and thermal studies were carried out to 675 °C. Remaining samples were all demagnetized in steps using thermal demagnetization procedures. Site mean statistics were calculated using a mean value for each core, combining thermal and AF measurements if necessary. Susceptibility measurements were made using a Sapphire Instruments susceptibility bridge. Isothermal remanent magnetization (IRM) measurements were made in fields up to 1.0 T using an ASC Impulse Magnetizer. Hysteresis properties were measured at room temperature in fields up to 2 T using a Princeton Applied Research micromagnetometer at the Institute for Rock Magnetism, University of Minnesota.

4.3 Rock-magnetic results

Samples from the Egersund–Ogna anorthosite show a range of responses to applied magnetic fields (Fig. 3). Sites known to have rare large multidomain magnetites from optical work (site 29) acquire nearly all of their saturation isothermal remanent magnetization (SIRM) by 0.1 T. Other sites, such as BK 56 and BK 68, have acquired approximately 50 per cent SIRM by 0.1 T and do not begin to approach saturation until above 0.5 T. Ratios of NRM to SIRM for the samples range from 0.9 to 2.7 per cent, indicating only a small amount of saturation in the natural state. Results from IRM acquisition indicate that magnetite, although present in some samples, is absent or only present in minor amounts in other samples.

In an attempt to understand further the magnetic phases present, hysteresis properties of individual samples were measured at room temperature. Samples showed a great range of values for both saturation magnetization (Ms) and saturation remanence (Mr). Two sample chips, one from a mega-orthopyroxene crystal and one from site BK 29, had large Ms values of 443 and 101 mA m^2 kg^{-1} respectively. The remaining 10 samples had much lower Ms values, ranging from 0.44 to 5.66 mA m^2 kg^{-1}. All samples had Mr/Ms ratios below 0.35 (Fig. 4), with no samples falling into the single domain (SD) field. Coercive force (Hc) values were more consistent, having a range from 3.14 to 43.9 mT, with the two lowest Hc values belonging to the mega-orthopyroxene grain and the sample from site BK 29, both which have optically visible magnetite. Coercivity of remanence (Hcr) measurements spanned from 5 to 174 mT. As seen in Fig. 4, the Hcr to Hc ratios are all between 1.4 and 5.5.

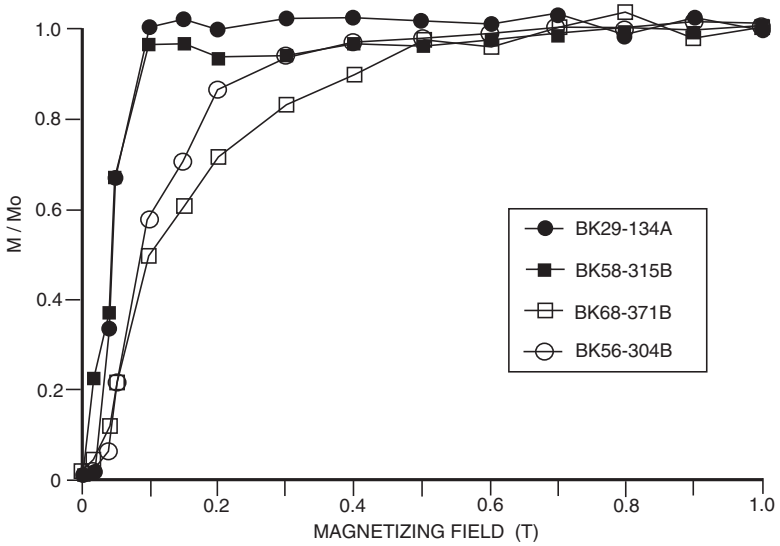


Figure 3. Plot of IRM normalized to magnetization at 1.0 tesla for four samples from the Egersund–Ogna anorthosite.

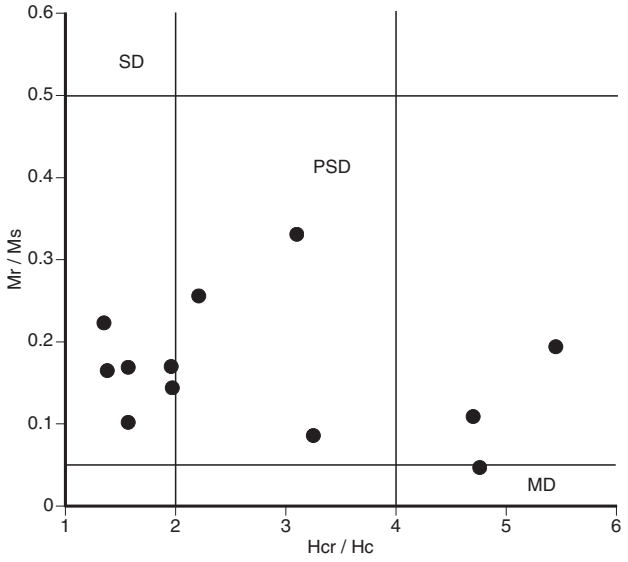


Figure 4. Plot of hysteresis parameter ratios for selected samples from the Egersund–Ogna anorthosite. The horizontal axis is the ratio of coercivity of remanence to coercive force. The vertical axis is ratio of saturation remanence to saturation magnetization. Thin lines divide fields for single-domain (SD), psuedo-single domain (PSD) and multidomain (MD) after Day *et al.* (1977), modified by Dunlop & Özdemir (1997). Note that domain divisions pertain to magnetite only.

Points falling towards the right-hand side of the graph represent samples with mixed mineralogy, having both haemo-ilmenite and (rare) magnetite grains. Samples to the left of the diagram fall beyond the pseudo-single domain range for magnetite grains (Day *et al.* 1977). The results here are similar to results for natural haemo-ilmenites studied by Dyar *et al.* (2004), and are in the region for results from haematite-dominated sedimentary and metamorphic rocks (Molina-Garza *et al.* 1996; McEnroe & Brown 2000a,b).

4.4 Palaeomagnetic results

There is a large range of initial NRM intensities from 1.544 down to 0.004 A m^{-1} (Table 1). Likewise, there is considerable variation in bulk susceptibility, with a mean value of 4.09×10^{-4} SI but ranging from a high of 22.4×10^{-4} SI to a low of 0.29×10^{-4} SI. The resulting Koenigsberger ratios (Q) for all sites are greater than 1, and nearly half are greater than 35, indicating that the remanent magnetization is stronger than the induced magnetization and overall the remanent component should dominate the magnetic response of the rock body.

Progressive demagnetization, either thermal or AF, reveals a relatively high magnetic stability and a consistent single-component magnetization. Samples showed high unblocking temperatures, with little or no loss of intensity until 525 to 550 °C (Fig. 5). The loss in intensity after 525 °C spreads over 75°, with only a few per cent remaining at 600 °C (Fig. 5a), or drops sharply from 550 °C to 5 per cent remaining at 575 °C (Fig. 5b). In some cases considerable intensity ($\sim$ 30 per cent) persists above 600 °C (Fig. 5c). Mean destructive temperatures are generally between 550° and 575 °C.

AF demagnetization shows a wide range of mean destructive fields (MDFs) from less than 10 mT (Fig. 5b) to 80 mT (Fig. 5c), with a majority of samples having MDFs of $\sim$ 40 mT. Decay curves can be concave upward, concave downward or L-shaped with loss of a large viscous component early in the demagnetization process.

As seen in Fig. 5, paired specimens from the same core show reasonably consistent behaviour and directions during both thermal and AF demagnetization. Most cores indicate some minor variations in directions at low demagnetization levels followed by decay to or near the origin. Thermal demagnetization appears to provide better directional information, especially in cases where large viscous components are present (Fig. 5b). All samples measured had steep negative inclinations and declinations biased to the northwest. Although minor overprints are removed at low demagnetization values, there is no indication of any positive inclination components.

Characteristic remanent directions were determined for each sample using line-fitting to the decay curves, with at least three points needed for a line and maximum angular deviation values averaging 5° or less. If more than one specimen per core was measured, directions were averaged, so each core had equal weight in the site mean calculations. Resulting directions and virtual geomagnetic poles for the 13 sites are listed in Table 1 and plotted in Fig. 6. The mean direction for the Egersund–Ogna anorthosite is $D = 325.^{\circ}9$, $I = -80.^{\circ}1$ and $a_{95} = 4.^{\circ}9$. The corresponding geomagnetic pole is at 42.°1S latitude and 200.°4E longitude, with an A_{95} of 9.°0.

Table 1. Palaeomagnetic data from the Egersund–Ogna anorthosite, Rogaland, Norway.

Site	N/No	χ 10^{-4} SI	Int. mA m^{-1}	Q	Inc.	Dec.	k	α_{95}	Pole Lat.	Pole Long.
BK 9	5/5	1.40	33	6	−75.8	304.1	251	4.8	−39.0	214.7
BK 29	5/5	22.4	1544	11	−70.9	316.7	243	4.9	−29.0	212.5
BK 30	4/4	0.89	280	70	−79.5	292.3	83	10.1	−46.9	214.2
BK 56	5/6	4.00	479	46	−79.2	300.9	81	8.5	−44.5	211.4
BK 57	5/5	7.12	551	16	−75.9	5.6	60	9.9	−32.1	183.1
BK 58	5/6	9.03	820	35	−67.5	350.8	204	5.4	−19.2	192.2
BK 59	6/6	0.38	4	3	−84.5	250.8	170	5.2	−60.4	207.3
BK 60	5/5	1.22	83	16	−81.2	239.0	248	4.9	−63.4	220.3
BK 61	7/7	0.47	42	16	−88.1	354.9	28	11.6	−54.7	186.6
BK 68	5/5	0.69	30	8	−76.6	306.5	773	2.8	−39.5	212.6
BK 69	6/6	0.29	230	148	−82.3	29.3	242	4.3	−44.8	175.7
BK 70	4/4	3.44	924	54	−72.2	10.9	156	7.4	−26.1	179.5
BK 71	5/5	1.78	393	36	−76.8	322.6	79	8.6	−36.5	204.8
MEAN	13/13				−80.1	325.9	73	4.9	−41.5	200.3
	13/13							9.0	−42.1	200.4

N/No, number of samples (sites) used in calculation/number of samples (sites) measured; χ, mean bulk susceptibility; Int., mean magnetic intensity at NRM level; Inc. < Dec., mean inclination and declination of site in degrees; k, precision parameter; α_{95}, 95 per cent confidence circle; Pole Lat. and Long., mean geomagnetic pole in degrees south latitude and east longitude.

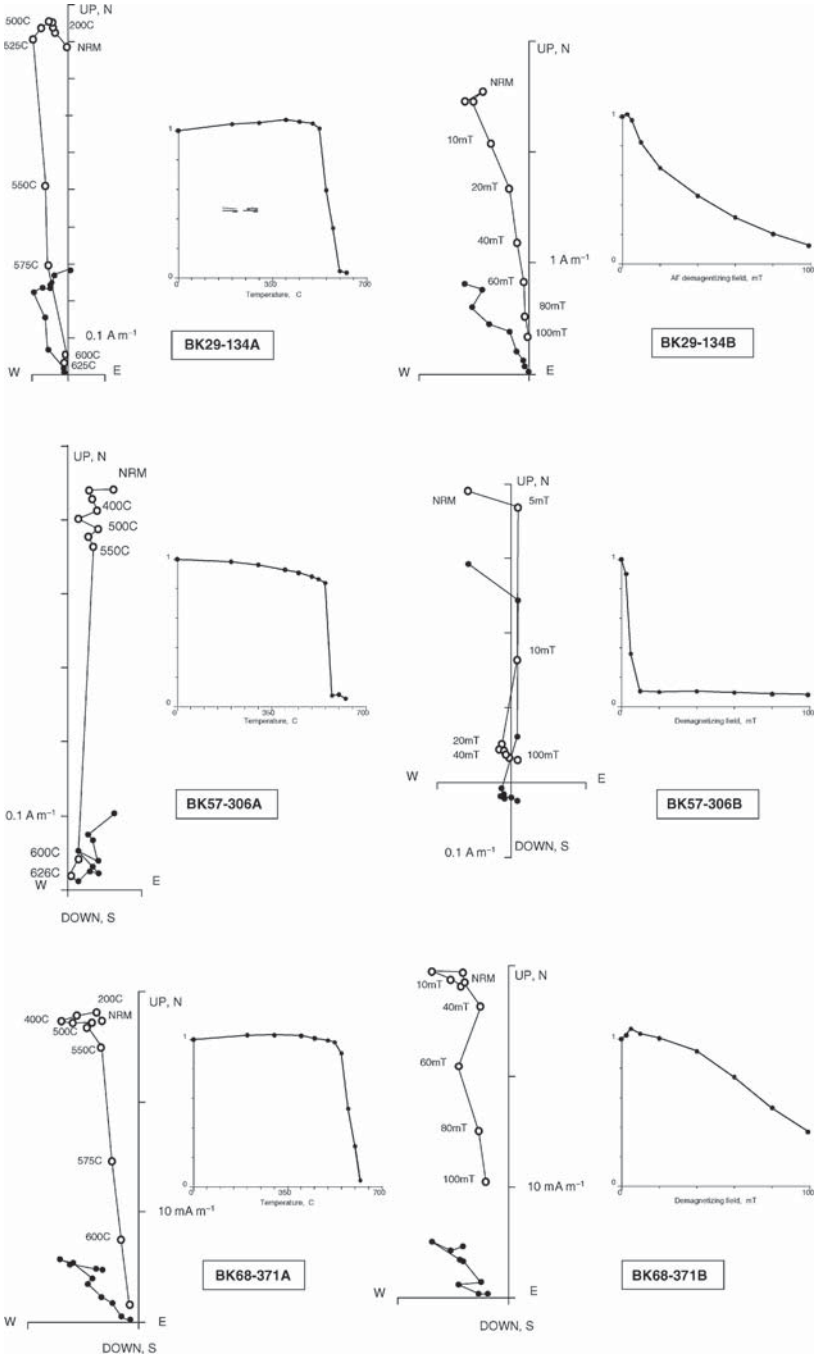


Figure 5. Vector end-point diagrams for alternating field and thermal demagnetization and relative intensity loss during demagnetization steps. Open circles are projections on to the vertical plane; solid circles represent projections on to the horizontal plane. Vertical components are all steep and up (negative); horizontal components range from northeast to southwest.

5 DISCUSSION

5.1 Age and nature of the magnetization

The rocks of the Egersund–Ogna anorthosite are characterized by low susceptibilities, low intensities, high Q-values and stable remanence. The magnetic behaviour, as well as optical observations, indicates that minor haemo-ilmenite and/or ilmeno-haematite are predominant carriers of remanence, consistent with the conclusion of Hargraves & Fish (1972) that titaniferous haematite intergrown with ilmenite was responsible for the observed magnetization. The high unblocking temperatures paired with high mean destructive fields in alternating field demagnetization and the low bulk susceptibility values all point to these oxide phases. Also supportive of haemo-ilmenite and little or no magnetite are the hysteresis properties and saturation behaviour. None of the hysteresis data plots in the SD magnetite field and only one sample, with known magnetite, falls in the multidomain (MD) region. Coercivity values for many

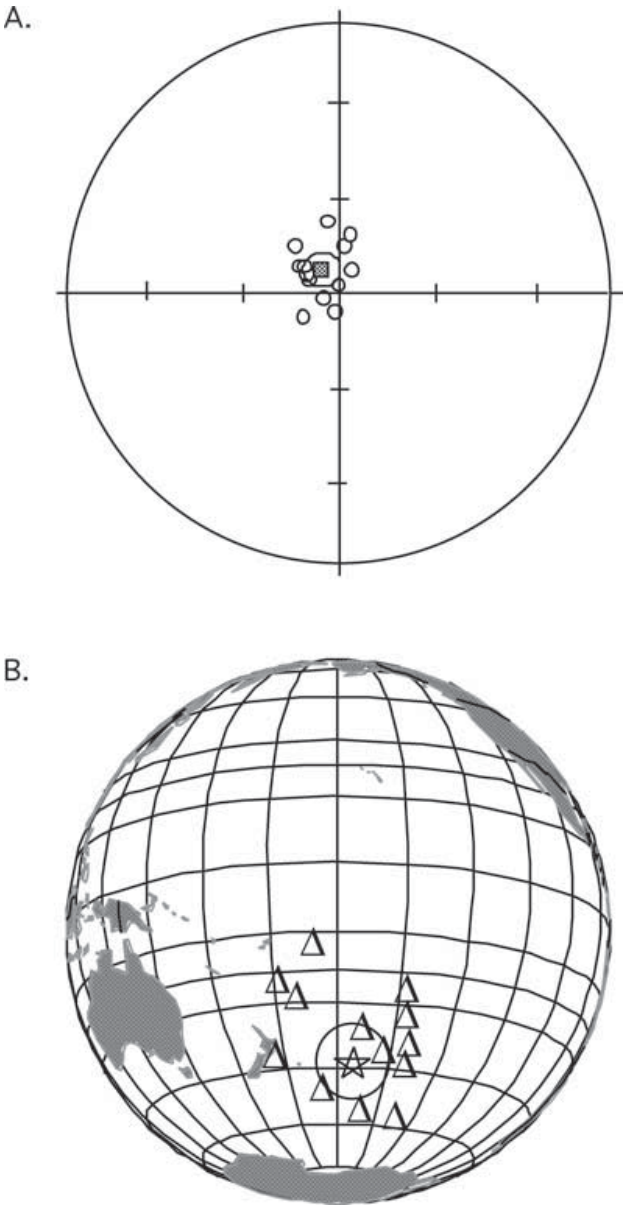


Figure 6. (a) Equal-area plot of site mean directions of all the Egersund–Ogna sites. Open circles are projections on to the upper hemisphere. The grey square is mean direction of the sites on the upper hemisphere, surrounded by the 95 per cent confidence circle. (b) Plot of virtual geomagnetic poles for the Egersund–Ogna sites. The star is mean geomagnetic pole for Egersund–Ogna.

of the samples have values higher than expected for magnetite. Although there is certainly magnetite present in some of the samples, it appears not to be SD in size and to occur only in minor amounts. Based on the susceptibility data, a maximum magnetite content of 0.01 per cent is calculated. Because some of the susceptibility is due to the haemo-ilmenite and ilmeno-haematite grains, the real 'magnetite content' is likely even lower.

These conclusions are supported by reflected-light microscopy where observations of magnetite grains of any size are rare, whereas haemo-ilmenite is common and ilmeno-haematite grains are subordinate. The moderate to high magnetic stability of the exsolved haematite–ilmenite intergrowths is similar to that reported from massive haemo-ilmenite layers in the adjacent Åna–Sira anorthosite (McEnroe *et al.* 2002), Sveconorwegian rocks from southern Sweden (McEnroe *et al.* 2001a) and the Grenville-age sillimanite gneisses from the Adirondacks of New York State (McEnroe & Brown 2000b).

Characteristic remanence directions are similar for all samples regardless of the specific oxide mineralogy, and all the phases observed appear to be primary igneous grains. This points to acquisition of the remanence during cooling soon after primary igneous crystallization at ∼930 Ma. U–Pb geochronology studies in the area indicate that there was a prolonged period of regional cooling from granulite-facies conditions at ∼ 980 Ma to below 610 °C at 918 ± 2 Ma, using titanite ages of Bingen & Van Breemen (1996), interrupted by the emplacement of these anorthosites and the Bjerkreim–Sokndal (BKS) layered intrusion. $^{40}Ar/^{39}Ar$ ages on hornblendes in the Rogaland area indicate regional cooling to below 550 °C by 916 ± 2 Ma (Bingen *et al.* 1998a). Thermal modelling of the Rogaland igneous complex by Westphal *et al.* (2003) indicates that the initial emplacement of the anorthosite produced a regional thermal gradient ranging from 640 °C to ∼ 880 °C. This was punctuated by a strong thermal pulse caused by the emplacement of the BKS layered intrusion. Based on age data and the model, the intrusive magmatic activity occurred in a short duration between ∼ 932 and ∼ 926 Ma. Best estimates using their cooling model suggest that the Egersund–Ogna anorthosite cooled through 400 °C at ∼900 Ma (Schumacher, private communication). Using the most recent phase diagram for the haematite–ilmenite solid solution by Harrison (McEnroe *et al.* 2002) which uses the thermodynamic constraints of Ghiorso (1997), abundant exsolution of haematite and acquisition of remanent magnetization will occur slightly below ∼390 °C for the bulk compositions of the haemo-ilmenite in the Egersund–Ogna anorthosite. Based on the phase diagram and thermal models, our best estimate for the age of remanence is approximately 900 Ma.

The question of later remagnetization has been raised by Walderhaug *et al.* (1999). They obtained similar palaeomagnetic directions from the Hunnedalen dykes which are younger, with an $^{40}Ar/^{39}Ar$ age of 848 ± 27 Ma (Walderhaug *et al.* 1999) and an Sm–Nd age of 855 ± 51 Ma (Maijer & Verschure 1998). Walderhaug *et al.* (1999) considered magnetite to be the predominant magnetic remanence carrier in the region, and suggested that a post-900 Ma metamorphic or hydrothermal event might have remagnetized the entire region. Although local areas of alteration are noted, none of the sites collected in Egersund–Ogna rocks shows signs of a pervasive later remagnetization event, either in demagnetization studies or in optical observations. Furthermore, we do not find evidence for a pervasive fluid event in our samples, nor is this observed in samples from the adjoining BKS intrusion (McEnroe *et al.* 1996, 2000, 2001b). Detailed petrologic studies on the layered intrusion (Wilson *et al.* 1996; Robins & Wilson 2001) indicate minor alteration, even to olivine, a mineral highly susceptible to alteration. The lack of pervasive alteration to the oxides in the sites sampled, in the haemo-ilmenite deposits (McEnroe *et al.* 2002) and in the BKS intrusion (McEnroe *et al.* 2004), supports the contention that the characteristic remanence was acquired during original cooling from high temperature. If the oxides were to be reset by a 'dry metamorphic' event, little evidence would be seen in the texture of the oxides; however, temperatures would have to have exceeded the unblocking temperatures of the haematite lamellae within the oxide grains. Owing to the single-domain size of the lamellae, these temperatures would have to have exceeded 500 °C, and evidence is lacking for such a high-temperature regional metamorphic event

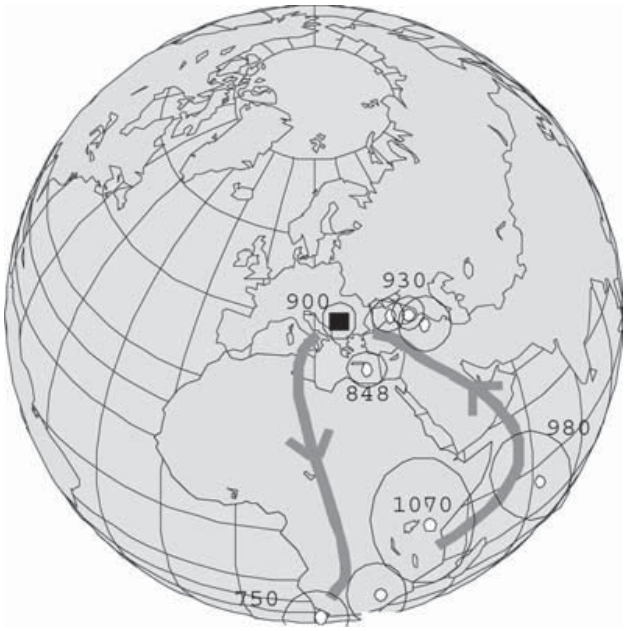


Figure 7. APWP for Baltica during the late Proterozoic (∼ 1100 to ∼ 750 Ma). Poles are listed in Table 2; for each study the south pole is shown with the best estimate of age and surrounded by its circle of 95 per cent confidence. New results from the Egersund–Ogna anorthosite are shown by the solid black square.

(Bingen & Van Breemen 1996; Bingen *et al.* 1998a). In addition, oxygen and carbon isotopes preclude pervasive regional passage of significant quantities of isotopically homogenous oxygen-bearing metamorphic fluids, in agreement with petrologic observations in the region (Bol *et al.* 1995). No evidence exists to indicate that this body has been tilted or deformed after intrusion and cooling, and there currently is no way to perform a tilt test. As directions from the other intrusions in the Rogaland complex have similar palaeomagnetic directions to the Egersund–Ogna (Stearn & Piper 1984), it is unlikely that this one body moved independently. Tilting of the entire region of southern Norway since Sveconorwegian time is a possibility, but one for which there is no evidence nor any conceivable test at the present time.

5.2 Apparent polar wander for Baltica

Shown in Fig. 7 is an apparent polar wander path for Baltica for part of the late Proterozoic (∼ 1100 to 750 Ma). The IAGA database has a number of entries for this time period, but most are not suitable for constructing polar wander paths. The most severe problem with these studies is the lack of well-constrained independent age control. Many studies are 'dated' only to the most general degree, and often an age is determined initially by the 'fit' of the palaeomagnetic data to the existing polar wander curve. These same studies are then later used, with the assigned ages, to reconstruct the pole path. Even when rocks do have modern radiometric age control, they often suffer from large error bars, spanning 100 Ma, or no case has been made in the literature to indicate the relationship of the age of magnetization to the radiometric age. Additional problems are encountered with a number of studies in which only a few sites and/or samples have been measured, severely jeopardizing the reliability of the results.

When strict age criteria are applied, the number of acceptable poles dwindles considerably to those listed in Table 2 and plotted in Fig. 7. Note that poles are plotted in or near the Northern Hemisphere in what has been assumed to be the South Pole position. The path appears to be a large loop, long recognized in the literature (e.g. Elming *et al.* 1993), showing counter-clockwise motion in the Northern Hemisphere. Poles prior to 930 Ma are not well defined, and although there are acceptable ages on several poles of Sveconorwegian time, the poles themselves are severely limited by low numbers of palaeomagnetic sites.

The magnetic pole for the Egersund–Ogna anorthosite from this study shows reasonable agreement with earlier poles calculated for Rogaland rocks.. The poles form a swath centred on 40° N latitude and stretching from 20° to 40° E longitude (Fig. 7). From what is currently known about Rogaland rocks, there does not seem to be any relationship between pole position and perceived age of the units, although earlier workers were not able to consider the age of remanence acquisition compared with isotopic ages. The observed spread of poles may be due to actual differences in the timing of magnetization within the numerous Rogaland bodies as argued by Stearn & Piper (1984), to minor undetected overprints in some of the earlier studies, particularly those with inadequate demagneion, or to undetected local differential tilts. It is evident that Baltica was at high latitude at 930–900 Ma, as has been suggested by earlier studies (Elming *et al.* 1993; Weil *et al.* 1998).

Also included in Fig. 7 are two studies from the Hunnedalen dykes, the earlier study by Poorter (1972) and the more recent data from Walderhaug *et al.* (1999). Radiometric dating on these dykes gives an age of 848 ± 27 Ma using $^{40}Ar/^{39}Ar$ and 855 ± 59 Ma using Sm–Nd (Walderhaug *et al.* 1999). Both of these studies fall close to the poles from Rogaland, although their magnetizations are younger. If this is correct, then although considerable motion of Baltica occurred between 970 and 900 Ma, there was much less motion of the craton in the next 70 Myr (930 to 850 Ma). The next younger poles available are from northernmost Scandinavia and have an age of ∼ 750 Ma, although this is not well determined.

Table 2. Selected palaeomagnetic pole positions (South poles) for Baltica, 1100–750 Ma.

Study	Age	Sites	Pole	A_{95}	Reference
Sredny–Kildin Red Beds	∼ 715	7	−31/16	7	Shipunov & Chumakov (1991)
Vadsø Group	∼ 750	11	−23/29	9	Bylund (1994)
Hunnedalen dykes	848 ± 27	8	34/28	5	Poorter (1972)
Hunnedalen dykes	848 ± 27	6	41/42	6	Walderhaug *et al.* (1999)
Egersund–Ogna	900	16	42/20	4	Hargraves & Fish (1972) and this study
Other Rogaland anorthosites	930	42	44/34	3	Mean of data in Stearn & Piper (1984)
Bjerkreim–Sokndal intrusion	930	36	44/38	4	Mean of data in Stearn & Piper (1984)
Swedish dolerites	∼ 980	7	−7/61	11	Mean of dykes in Patchett & Bylund (1977)
Bramble Mean	1100–1040	4*	−3/37	15	Meert & Torsvik (2003)

Age is in millions of years; Pole, latitude and longitude of south geomagnetic pole; A_{95}, 95 per cent confidence circle about the mean pole; *number of separate studies.

If we accept these two studies, the pole position for Baltica returns to equatorial, not far from the early 1070-Ma position. Although the APWP shown in Fig. 7 is counter-clockwise, the limited number of poles makes this distinction tentative at best.

The position of Baltica within the supercontinent Rodinia has seen a number of likely scenarios, from tight connections between western Baltica and eastern North America (Dalziel 1991) or Baltica and South America (Park 1992), to displacements far to the north (Karlstrom *et al.* 2001) or upside down (Hartz & Torsvik 2002; Torsvik 2003). In recent stringent pole selections, producing 'key' poles for cratons (Buchan *et al.* 2001), there are no acceptable poles for Baltica in the 1100- to 600-Ma range, making comparisons of relative continental positions difficult. Although the magnetization defining the pole for the Egersund–Ogna anorthosite presented here is well dated, there are few if any similar age poles in North America for comparison.

6 CONCLUSIONS

The palaeomagnetic study of rocks from the Egersund–Ogna anorthosite in the Rogaland igneous complex shows that these samples contain a stable remanent magnetization with steep negative inclinations and predominantly northwestern declinations. Demagnetization studies, as well as optical observations, susceptibility values and rock magnetic data, indicate that this stable remanence is probably carried by haemo-ilmenite and/or ilmeno-haematite phases. Low susceptibility values but relatively high remanence values produce high Q-ratios for these rocks. Lack of pervasive alteration in the rocks indicates that remagnetization is unlikely. The characteristic remanence, identified in both AF and thermal demagnetization, is taken to represent an age younger than the intrusion age of 930 Ma, and likely to be near the hornblende ages of 916 ± 2 Ma representing regional cooling to below 550 °C. Our best estimate for remanence acquisition is near to 900 Ma. Mean palaeomagnetic poles from the Egersund–Ogna body and from other intrusive bodies in the Rogaland igneous complex all suggest that Fennoscandia was at high (southern) latitudes during this time. Although data from Rogaland are well constrained magnetically and geochronologically, considerable caution needs to be applied to interpreting other parts of the APWP.

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REFERENCES

Barnichon, J.D., Havenith, H., Hoffer, B., Charlier, R., Jongmans, D. & Duchesne, J.C., 1999. The deformation of the Egersund–Ogna anorthosite massif, south Norway: finite-element modeling of diapirism, *Tectonophys.,* **303,** 109–130.

Bingen, B. & Van Breemen, O., 1996. U-Pb Titanite geochronology in Rogaland-Vest Agder (SW Norway): Regional temperature at the time of intrusion of the Rogaland anorthosites, in *Petrology and Geochemistry of magmatic suites of rocks in continental and oceanic crust,* pp. 145–159, ed. Demaiffe, D., Université Libre de Bruxelles, Brussels.

Bingen, B., Boven, A., Punzalan, L., Wijbrans, J. & Demaiffe, D., 1998a. Hornblende Ar/Ar geochronology across terrane boundaries in the Sveconorwegian Province of S. Norway, *Precambrian Res.,* **90,** 159–185.

Bingen, B., Demaiffe, D. & van Breemen, O., 1998b. The 616 Ma Egersund basaltic dike swarm, SW Norway, and Late Neoproterozoic opening of the Iapetus Ocean, *J. Geol.,* **106,** 565–574.

Bol, L.C.C., Nijland, T.G., Sauter, P.C.C., Jansens, J.B.H. & Valley, J.W., 1995. Preservation of premetamorphic oxygen and carbon isotopic trends in granulite facies marble from Rogaland, southwest Norway, *Am. J. Sci.,* **295,** 1179–1219.

Buchan, K., Mertanen, S., Park, R., Pesonen, L., Elming, S.-A., Abrahamsen, N. & Bylund, G., 2000. Comparing the drift of Laurentia and Baltica in the Proterozoic: the importance of key paleomagnetic poles, *Tectonophys.,* **319,** 167–198.

Buchan, K., Ernst, R.E., Hamilton, M.A., Mertanen, S., Pesonen, L.J. & Elming, S.-Å., 2001. Rodinia: the evidence from integrated paleomagnetism and U-Pb geochronology, *Precambrian Res.,* **110,** 9–32.

Bylund, G., 1994. Paleomagnetism of the Late Precambrian Vadso and Barents Sea Groups, Varanger Peninsula, Norway, *Precambrian Res.,* **69,** 81–93.

Dalziel, I., 1991. Pacific margins of Laurentia and east Antarctica-Australia as a conjugate rift pair: Evidence and implications for an Eocambrian supercontinent, *Geology,* **19,** 598–601.

Day, R., Fuller, M.D. & Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: Grain size and composition dependence, *Phys. Earth planet. Inter.,* **13,** 260–267.

Duchesne, J.-C. (ed.), 2001. The Rogaland Intrusive Massifs, an excursion guide, *Norges Geologiske Undersokelse,* Report 2001.29, 137pp.

Duchesne, J.-C. & Maquil, R., 1987. The Egersund–Ogna massif, The Geology of Southernmost Norway, in *Norges Geologiske Undersokelse,* Vol. 1, pp. 50–56, eds Maijer, C. & Padget, P., NGU, Trondheim, Norway, Special Publication.

Dunlop, D.J. & Özdemir, Ö, 1997. *Rock Magnetism: Fundamentals and Frontiers,* Cambridge Univ. Press, Cambridge, 573pp.

Dyar, M.D., McEnroe, S.A., Murad, E., Brown, L.L. & Schiellerup, H., 2004. The relationship between exsolution and magnetic properties in hemo-ilmenite: Insights from Mössbauer spectroscopy with implications for planetary magnetic anomalies, *Geophys. Res. Lett.,* **31,** doi:10.1029/2003/GL019076, L04608.

Ghiorso, M.S., 1997. Thermodynamic analysis of the effect of magnetic ordering on miscibility gaps in the Fe-Ti cubic and rhombohedral oxide minerals and the FeTi oxide geothermometer, *Phys. Chem. Miner.,* **25,** 23–38.

Elming, S. *et al*., 1993. The drift of the Fennoscandian and Ukrainian shields during the Precambrian: a paleomagnetic analysis, *Tectonophys.,* **223,** 177–198.

Hargraves, R. & Fish, J., 1972. Paleomagnetism of Anorthosite in Southern Norway and Comparison with an Equivalent in Quebec, *Geol. Soc. Am.,* Memoir **132,** 141–148.

Hartz, E. & Torsvik, T., 2002. Baltica upside down: A new plate tectonic model for Rodinia and the Iapetus Ocean, *Geology,* **30,** 255–258.

Hoffman, P., 1991. Did the Breakout of Laurentia Turn Gondwanaland Inside-Out?, *Science,* **252,** 1409–1412.

Karlstrom, K.E., Åhäll, K.-I., Harlan, S.S., Williams, M.L., McLelland, J. & Geissman, J.W., 2001. Long-lived (1.8–1.0 Ga) convergent orogen in southern Laurentia, its extensions to Australia and Baltica, and implications for refining Rodinia, *Precambrian Res.,* **111,** 5–30.

Korneliussen, A., Nilsson, L.P., Schiellerup, H., Gautneb, E. & McEnroe, S.A., 1999. Titanium deposits in Norway, in *Mineral Deposits: Processes to Processing,* pp. 1137–1140, eds Stanley C.J. *et al*., A. A. Balkema, Rotterdam.

McEnroe, S.A. & Brown, L.L., 2000a. Palaeomagnetism, rock magnetism, and geochemistry of Jurassic dykes and correlative redbeds, Massachusetts, USA, *Geophys. J. Int.,* **143,** 22–38.

McEnroe, S.A. & Brown, L.L., 2000b. A closer look at remanence-dominated anomalies: Rock-magnetic properties and magnetic mineralogy of the Russell Belt microcline-sillimanite gneisses, Northwest Adirondack Mountains, New York, *J. geophys. Res.,* **105,** 16 437–16 456.

McEnroe, S.A., Robinson, P. & Panish, P.T., 1996. Rock magnetic properties, oxide mineralogy, and mineral chemistry in relation to aeromagnetic interpretation and search for ilmenite reserves, *NGU Rep.,* **96 060,** 153pp

McEnroe, S.A., Robinson, P. & Panish, P., 2000. Chemical and petrographic characterization of ilmenite- and magnetic-rich cumulates of the Sokndal region, Rogaland, Norway, *Norwegian Geol. Survey Bull.,* **426,** 49–56.

McEnroe, S.A., Harrison, R., Robinson, P., Golla, U. & Jercinovic M., 2001a. Effect of fine-scale microstructures in titanohaematite on the acquisition and stability of natural remanent magnetization in granulite facies metamorphic rocks, southwest Sweden: Implications for crustal magnetism, *J. geophys. Res.,* **106,** 30 523–30 546.

McEnroe, S.A., Robinson, P. & Panish, P., 2001b. Aeromagnetic anomalies, magnetic petrology and rock magnetism of hemo-ilmenite- and magnetite-rich cumulates from the Sokndal Region, South Rogaland, Norway, *Am. Mineral.,* **86,** 1447–1468.

McEnroe, S.A., Harrison, R.J., Robinson, P. & Langhorst, F., 2002. Nanoscale haematite-ilmenite lamellae in massive ilmenite rock: an example of 'Lamellar Magnetism' with implications for planetary magnetic anomalies, *Geophys. J. Int.,* **151,** 890–912.

McEnroe, S.A., Brown L.L. & Robinson, P., 2004. Earth analog for Martian magnetic anomalies: Remanence properties of hemo-ilmenite norites in the Bjerkreim-Sokndal Intrusion, Rogaland, Norway, *J. appl Geophys.,* in press.

Maijer, C. & Verschure, R., 1998. Petrology and isotope geology of the Hunnedalen Monzonoritic dyke swarm, SW Norway: a possible late expression of Egersund Anorthosite magmatism, *Norges Geol. Undersokelse,* Bulletin **434,** 83–107.

Marker, M., Schiellerup, H., Meyer, G., Robins, B. & Bolle, O., 2003a. Geological map of the Rogaland Anorthosite Province, 1:75,000, *Norges Geol. Undersokelse,* Special Publication **9,** Plate 1.

Marker, M., Schiellerup, H., Meyer, G., Robins, B. & Bolle, O., 2003b. Introduction to the geological map of the Rogaland Anorthosite Province 1:75,000, *Norges Geol. Undersokelse,* Special Publication **9,** 109–116.

Meert, J.G. & Powell, C.M.A., 2001. Assembly and break-up of Rodinia: introduction to the special volume, *Precambrian Res.,* **110,** 1–8.

Meert, J.G. & Torsvik, T.H., 2003. The making and unmaking of a supercontinent: Rodinia revisited, *Tectonophys.,* **375,** 261–288.

Michot, J. & Michot, P., 1969. The Problem of Anorthosites: The South-Rogaland Igneous Complex, Southwestern Norway, in *Origin of Anorthosite and Related Rocks,* Vol. 18, pp. 399–410, ed. Isachsen, Y.W., Univ. State New York, Alloany, NY.

Michot, P., 1969. Geological Environments of the Anorthosites of South Rogaland, Norway, in *Origin of Anorthosite and Related Rocks,* Vol. 18, pp. 411–423, ed. Isachsen, Y.W. , Univ. State New York, Alloany, NY.

Molina-Garza, R.S., Geissman, J.W., Lucas, S.G. & Van der Voo, R., 1996. Paleomagnetism and magnetostratigraphy of Triassic strata in the Sangre de Cristo Mountains and Tucumcari Basin, New Mexico, USA, *Geophys. J. Int.,* **124,** 935–953.

Oftedahl, C., 1980. Geology of Norway, *Norges Geol. Undersokelse,* Bulletin **54,** 3–114.

Park, R.G., 1992. Plate kinematic history of Baltica during the Middle to Late Proterozoic: A model, *Geology,* **20,** 725–728.

Patchett, P.J. & Bylund, G., 1977. Age of Grenville Belt magnetization Rb-sr and paleomagnetic evidence from Swedish dolerites, *Earth Planet. Sci. Lett.,* **35,** 92–104.

Pesonen, L., Torsvik, T., Elming, S. & Bylund, G., 1989. Crustal evolution of Fennoscandia—paleomagnetic constraints, *Tectonophys.,* **162,** 27–49.

Pesonen, L., Bylund, G., Torsvik, T., Elming, S. & Mertanen, S., 1991. Catalogue of paleomagnetic directions and poles from Fennoscandia: Archean to Tertiary, *Tectonophys.,* **195,** 151–207.

Piper, J., 1987. *Paleomagnetism and the Continental Crust,* Open University Press, New York.

Poorter, R., 1972. Paleomagnetism of the Rogaland Precambrian (Southwest Norway), *Phys. Earth planet. Inter.,* **5,** 167–176.

Robins, B. & Wilson, J.R., 2001. The Bjerkreim-Sokndal layered intrusion, in *The Rogaland Intrusive Massive an excursion guide,* pp. 35–47, ed. Duchesne, J.C., NGU Report 2001.029, Trondheim, Norway.

Robinson, P., Harrison, R.J., McEnroe, S.A. & Hargraves, R., 2002. Lamellar magnetism in the haematite-ilmenite series as an explanation for strong remanent magnetization, *Nature,* **418,** 517–520.

Robinson, P., Harrison, R.J., McEnroe, S.A. & Hargraves, R.B., 2004. Nature and origin of lamellar magnetism in the haematite-ilmenite series, *Am. Min.,* **89,** 725–747.

Schärer, U., Wilmart, E. & Duchesne, J., 1996. The short duration and anorogenic character of anorthosite magmatism: U-Pb dating of the Rogaland complex, Norway, *Earth planet. Sci. Lett.,* **139,** 335–350.

Shipunov, S.V. & Churnakov, N.M., 1997. Paleomagnetism of Upper Proterozoic deposits of the Kola Peninsula, *Geotectonics,* **25,** 401–410.

Starmer, I., 1996. Accretion, rifting, rotation and collision in the North American supercontinent, 1700–950 Ma, in *Precambrian crustal Evolution in the North Atlantic Region,* Vol. 112, Geological Society Special Publication, pp. 219–248, ed. Brewer, T., Geological Society, London.

Stearn, J. & Piper, J., 1984. Paleomagnetism of the Sveconorwegian Mobile Belt of the Fennoscandian Shield, *Precambrian Res.,* **23,** 201–246.

Torsvik, T., 2003. The Rodinia jigsaw puzzle, *Science,* **300,** 1379–1381.

Torsvik, T., Smethurst, M., Meert, J., Van der Voo, R., McKerrow, W., Brasier, M., Sturt, B. & Walderhaug, H., 1996. Continental break-up and collision in the Neoproterozoic and Paleozoic—A tale of Baltica and Laurentia, *Earth-Science Rev.,* **40,** 229–258.

Van der Voo, R. & Meert, J., 1991. Late Proterozoic paleomagnetism and tectonic models: a critical appraisal, *Precambrian Res.,* **53,** 149–163.

Verschure, R.H., Andriesen, P.A.M., Boelrijk, N.A., Hebeda, E.H., Maijer, C., Priem, H.N.A. & Verdurmen, E.A., 1980. On the thermal stability of Rb-Sr and K-Ar biotite systems: Evidence from coexisting Sveconorwegian (ca. 870 Ma) and Caledonian (ca. 400 Ma) biotites in SW Norway:, *Contrib. Min. Petrol.,* **74,** 245–252.

Walderhaug, H., Torsvik, T., Eide, E., Sundvoll, B. & Bingen, B., 1999. Geochronology and paleomagnetism of the Hunnedalen dykes, SW Norway: implications for the Sveconorwegian apparent polar wander loop, *Earth planet. Sci. Lett.,* **169,** 71–83.

Weil, A., Van der Voo, R., Mac Niocaill, C. & Meert, J., 1998. The Proterozoic Supercontinent Rodinia: Paleomagnetically Derived Reconstructions for 1,100 to 800 Ma, *Earth planet. Sci. Lett.,* **154,** 13–24.

Westphal, M., Schumacher, J.C. & Boschert, S., 2003. High temperature metamorphism and the role of magmatic heat sources at the Rogaland Anorthosite Complex in southwest Norway, *J. Petrol.,* **44,** 1145–1162.

Wilson, J.R., Robins, B., Neilsen, F., Duchesne, J.C. & Vander Auwera, J., 1996. The Bjerkreim-Sokndal layered intrusion, southwest Norway, in *Layered Intrusions,* pp. 231–256, ed. Cawthorn, R.G., Elsevier, Amsterdam.